

Symmetry relations at phase transitions

15

15.1 Phase transitions in the solid state

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Definition 15.1 A phase transition is an event which entails a discontinuous (sudden) change of at least one property of a material.

Properties that can change discontinuously include volume, density, elasticity, electric, magnetic, optical, or chemical properties. A phase transition in the solid state is accompanied by a structural change, which means a change of crystallographic data for crystalline solids (space group, lattice parameters, occupied positions, atomic coordinates). In the literature, 'structural phase transitions' are indeed distinguished from 'magnetic', 'electronic', and other kinds of phase transitions, but all of these transitions are always accompanied by (sometimes very small) structural changes. For example, a transition from a paramagnetic to a ferromagnetic phase entails spontaneous magnetostriction, i.e. the structure experiences a slight deformation.

Even the so-called isostructural or isosymmetric phase transitions involve structural changes. These are transformations that exhibit no changes of the occupation of Wyckoff positions, the number of atoms in the unit cell, and the space group (within the scope of the parent clamping approximation; Section 11.1, page 137). For example, samarium sulfide, SmS, crystallizes in the NaCl type. At ambient pressure it is a black semiconductor. Under pressure it experiences a phase transition to a metallic, golden lustrous modification, combined with a sudden volume change by -14% and a delocalization of electrons from the $4f$ subshell of the samarium atoms into a metallic $5d6s$ band ('electronic phase transition': $\text{Sm}^{2+} + \text{S}^{2-} \rightarrow \text{Sm}^{3+} + \text{S}^{2-} e^-$). The golden SmS also has the NaCl structure; nevertheless, there is a structural change because the lattice parameter experiences a sudden decrease from $a = 591$ pm to $a = 562$ pm (values at 58 K and 1.13 GPa [263]).

There exist numerous kinds of phase transitions, and the corresponding theories and experimental findings are the subject of an extensive field of research in physics. This chapter gives only a small insight, group-theoretical aspects being in the foreground. A few additional considerations are dealt with in Appendix B (page 269). First, we introduce a few common distinctions.

15.1.1 First- and second-order phase transitions

¹ Monotropic phase transitions are irreversible, i.e. they start from a phase that is only kinetically stable, but thermodynamically unstable at any condition.

A thermodynamically stable phase can become unstable relative to another phase by a change of the external conditions (temperature, pressure, electric field, magnetic field, mechanical forces); this causes a stress that can induce a transformation. The transformation is enantiotropic, i.e. reversible; it can be reversed by returning to the original conditions.¹ If only one variable of state is changed, for example the temperature T or the pressure p , then there is a point of transition T_c or p_c at which the phases are at equilibrium with one another and where the Gibbs free energy change is $\Delta G = 0$.

Let: G = Gibbs free energy (free enthalpy), H = enthalpy, U = internal energy, S = entropy, and V = volume. They are functions of T and p and further variables of state like electric or magnetic fields; we restrict our considerations to the variables T and p .

For reversible processes, according to the laws of thermodynamics, (negative) entropy and volume are the first derivatives of the Gibbs free energy $G = H - TS = U + pV - TS$ with respect to temperature and pressure. The partial derivatives express that this is valid if the other variable(s) of state are held constant. The second derivatives with respect to T and p express the specific heat at constant pressure, $C_p = \partial H / \partial T$; the compressibility of the volume V at constant temperature, $\kappa V = -\partial V / \partial p$; and the thermal expansion of the volume, $\alpha V = \partial V / \partial T$, α being the coefficient of thermal expansion.

$$\frac{\partial G}{\partial T} = -S \quad \frac{\partial G}{\partial p} = V \quad (15.1)$$

$$\frac{\partial^2 G}{\partial T^2} = -\frac{\partial S}{\partial T} = -\frac{1}{T} \frac{\partial H}{\partial T} = -\frac{C_p}{T}$$

$$\frac{\partial^2 G}{\partial p^2} = \frac{\partial V}{\partial p} = -\kappa V$$

$$\frac{\partial^2 G}{\partial p \partial T} = \alpha V$$

Definition 15.2 after EHRENFEST (1933). At a **first-order phase transition** at least one of the first derivatives of the Gibbs free energy G experiences a discontinuous change, i.e. $\Delta S \neq 0$ or $\Delta V \neq 0$. It is accompanied by the exchange of conversion energy (latent heat) $\Delta H = T\Delta S$ with the surroundings. At a **second-order phase transition** volume and entropy experience a continuous variation, but at least one of the second derivatives of G exhibits a discontinuity. At a phase transition of n -th order, a discontinuity appears for the first time at the n -th derivative (however, third and higher order transitions do not really occur).

The mentioned distinction of the order of phase transitions after EHRENFEST is based on purely thermodynamic arguments and macroscopic measured variables, without taking into account the interatomic interactions and structures of the substances.

Since the times of EHRENFEST, theory and experimental measurement techniques have advanced considerably. The mentioned classification has turned out to be neither sufficiently ample nor sufficiently precise. In phase-transition physics, the distinction of discontinuous and continuous phase transitions has replaced the classification after EHRENFEST:

Definition 15.3 At a **discontinuous phase transition** entropy as well as an order parameter change discontinuously. At a **continuous phase transition** the entropy and order parameter change continuously (in infinitesimal small steps).

Section 15.2 explains what an order parameter is.

Often there is hardly any difference between the old and the new classification. The terms 'first' and 'second order' continue to be used. 'First-order transition' is often used as a synonym for 'discontinuous phase transition' since every discontinuous transition is also a first-order transition according to EHRENFEST. 'Second-order transition' usually means 'continuous phase transition'. In any case, all phase transitions, including the continuous transitions, exhibit a discontinuous behaviour of certain thermodynamic functions.

In the theory of continuous phase transitions, critical points are of essential importance, and the corresponding physical laws are called *critical phenomena*. Continuous phase transitions exhibit a number of properties in common ('universality'), in contrast to the discontinuous transitions. In particular, they do not depend on the kind of interaction between the atoms, but on their range of action and on the number of space dimensions in which they are active. For more details see Appendix B (page 269).

A discontinuous phase transition is always accompanied by an exchange of latent heat $\Delta H = T\Delta S$ with the surroundings. This exchange cannot occur instantly, i.e. with an infinitely fast transfer of heat. Accordingly, discontinuous phase transitions exhibit *hysteresis*: The transition lags behind the causing temperature or pressure change. When the point of transition is reached, nothing happens; under equilibrium conditions the transition does not get started. Only after the point of transition has been crossed, i.e. under non-equilibrium conditions, does the transition get started, provided that nuclei of the new phase are being formed which then grow at the expense of the old phase. Simultaneously, a temperature gradient is being built up with the surroundings, thus enabling the flow of latent heat. Predominantly, nuclei are formed at defect sites in the crystal.

In contrast, continuous phase transitions show no latent heat, no hysteresis, and no occurrence of metastable phases.

15.1.2 Structural classification of phase transitions

A classification of phase transitions frequently encountered in the literature is due to BUEGER [37]:

- (1) **Reconstructive phase transitions:** Chemical bonds are broken and re-joined; the reconstruction involves considerable atomic motions. Such conversions are always first-order transitions.
- (2) **Displacive phase transitions:** Atoms experience small shifts.
- (3) **Order-misorder transitions:**² Different kinds of atoms that statistically occupy the same crystallographic point orbit in a crystal become ordered in different orbits or vice versa. Or molecules that statistically take several orientations become ordered in one orientation.

The mentioned classification is often made according to the known structures before and after the phase transition, without experimental evidence of the actually occurring processes and atomic motions during the transition. Nevertheless, frequently there are no doubts as to what category a phase transition is to be assigned. However, the qualitative kind of the classification does not always permit a clear assignment. There exists no unanimous opinion in

²The common term is term 'order-disorder transition'. However, we take the liberty of replacing the unfortunate term 'disorder' by the more precise term 'misorder' because there is still some order in the 'disordered' state; it is merely an order with faults.

the literature, of what has to be considered to be a displacive transition and how it should be delimited on the one hand from reconstructive, on the other hand from order–misorder transitions. For example, some authors call transitions displacive only if the atomic shifts are not only small, but also take place continuously. Some other authors call them displacive if there is a group–subgroup relation between the involved space groups and reconstructive if not.

On the other hand, BUERGER's classification sometimes permits subtle distinctions. For example, PbTiO_3 and BaTiO_3 both crystallize at high temperatures in the cubic perovskite type ($Pm\bar{3}m$). PbTiO_3 has a Ti atom at the centre of its coordination octahedron. In BaTiO_3 , however, it does not seem to be placed at the octahedron centre, but at eight positions slightly sideways from the centre, with occupancy probabilities of $\frac{1}{8}$ each. At the cubic $\rightarrow$ tetragonal transition the Ti atoms of PbTiO_3 slightly shift away from the centres of the octahedra, parallel to $+c$; that is a displacive transition. However, upon cooling, BaTiO_3 becomes tetragonal in that a Ti atom becomes more ordered by occupying only four out of the eight positions with occupancy probabilities of $\frac{1}{4}$ each. Upon further cooling it restricts itself to two positions (orthorhombic) and finally to one position (rhombohedral). These are order–misorder transitions [264–266].

15.2 On the theory of phase transitions

15.2.1 Lattice vibrations

Lattice vibrations play a major role at phase transitions in the crystalline state. In physics, lattice vibrations are treated as quasiparticles, as *phonons*, by analogy to the photons of light. A vibration with a certain frequency corresponds to phonons of a certain energy. We do not deal here with the theory of vibrations, but mention only some terms that are important for the understanding of the following [267–271].

In solid-state physics a *mode* (vibrational mode) is a collective and correlated vibrational motion of atoms having certain symmetry properties.

The symmetry of point and space groups does not deal with time. The symmetry of time-dependent functions like vibrations is not covered. *Representation theory* offers the necessary tools to deal with the symmetry properties of vibrating atoms in molecules and crystals; it is not a topic of this book, but there exists plenty of literature (e.g. [15–22]). *Irreducible representations*³ (symmetry species) are used to designate the symmetry of vibrations. The symbols introduced by PLACZEK are commonly used to designate the symmetry of molecular vibrations [272, 273] (called Mulliken symbols in American literature because MULLIKEN adopted them to designate the symmetry of wave functions [274]). A list with the most important symbols can be found in Appendix C (page 279). The symbols are also suited for the designation of the symmetry of lattice vibrations in crystals at the Γ point of the Brillouin zone.

In the language of solid-state physics, the Brillouin zone is a polyhedron around the origin of a coordinate system in ' k space'.⁴ The origin of the Brillouin zone is called the Γ point. Every lattice vibration is represented by a point with coordinates k_1, k_2, k_3 in the Brillouin zone.

³ often called irrep, for short

⁴ k space corresponds to the reciprocal space common in crystallography, but is expanded by a factor of 2π .

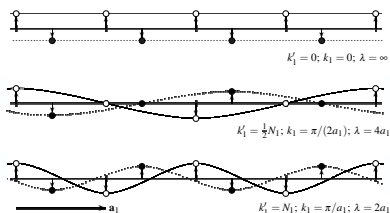


Fig. 15.1 Atomic displacements for three lattice modes of a chain consisting of two kinds of atoms. Other lattice modes with $0 < k_1 < \pi/a_1$ have wavelengths λ between $2a_1$ and ∞ . The number of nodes is $k'_1 = k_1 N_1 a_1 / \pi$ with N_1 = number of unit cells in the chain, a_1 = periodicity length (lattice constant); $0 \leq k'_1 \leq N_1$.

Three lattice vibrations are shown in Fig. 15.1 for the simple case of a chain (= crystal with one-dimensional translation symmetry) consisting of two kinds of atoms. N_1 is the number of unit cells in the chain of finite length, a_1 is the lattice constant, and k'_1 is the number of vibrational nodes of the standing wave. The standing wave with $k'_1 = 0$ has no nodes; all translation-equivalent atoms move in the same way and synchronously in all unit cells (in other words: the particle lattice of one kind of atoms vibrates relative to the particle lattice of the other kind of atoms). The vibration with $k'_1 = \frac{1}{2}N_1$ has half as many nodes as unit cells. The vibration with $k'_1 = N_1$ has exactly one node per unit cell; translation-equivalent atoms in adjacent cells move exactly in opposite directions (antisymmetric; opposite in phase). There can be no more nodes than unit cells. Instead of designating the number of nodes by integral numbers k'_1 that run from zero to N_1 , numbers $k_1 = k'_1 \pi / (N_1 a_1)$ are used which are independent of the number of cells. The maximal possible number of nodes $k'_1 = N_1$ then corresponds to the value $k_1 = \pi/a_1$. Three numbers $0 \leq k_1 \leq \pi/a_1$, $0 \leq k_2 \leq \pi/a_2$, $0 \leq k_3 \leq \pi/a_3$ are needed for a crystal with translational symmetry in three dimensions, and nodal surfaces replace nodal points.

A point at the boundary of the Brillouin zone has $k_i = \pi/a_i$. Special points at the Brillouin zone boundary are designated with capital latin letters (e.g. K, M, X). Vibrational modes at the Γ point, $k_1, k_2, k_3 = 0$, have no nodal surfaces; they have atoms that perform the same vibrational motions synchronously in all unit cells of the crystal. Only modes that are very close to the Γ point can be measured by optical techniques (infrared and Raman spectroscopy). Modes outside of the Γ point can be measured by inelastic neutron scattering.

For the example of CaCl_2 , one vibrational mode of the rutile-type modification is marked by arrows in Fig. 15.2 (top). This mode involves a mutual rotational vibration of the strands of edge-sharing coordination octahedra that run along c ; the motion is the same and synchronous in all unit cells and is thus at the Γ point. In tetragonal CaCl_2 ($P4_2/m2_1/n2/m$) this mode is of symmetry species B_{1g} . As explained in Table C.1 (page 279), in this case B means antisymmetric with respect to the 4_2 axes, i.e. after execution of a 4_2 operation the

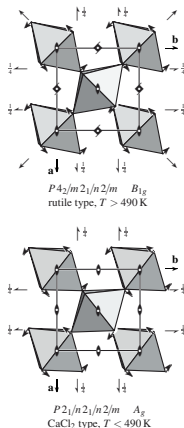


Fig. 15.2 Mutual rotations of the coordination octahedra of the soft mode of CaCl_2 in the rutile and CaCl_2 types.

⁵ g from the German *gerade*, meaning even.

direction of motion of the atoms has to be reversed. The lower-case 1 means symmetric with respect to the 2_1 axes parallel to **a** (and **b**). The index g means symmetric with respect to the points of inversion.⁵

In the orthorhombic low-temperature modification of CaCl_2 ($P2_1/n2_1/n2/m$; Fig. 15.2, bottom), the same kind of mode is symmetric with respect to all symmetry elements, it is *totally symmetric* (also called the *identity* or *unit representation*). Totally symmetric means that the space-group symmetry is completely fulfilled at any time of the vibrational motion. The symmetry species is A_g . The B_{1g} mode of the tetragonal as well as the A_g mode of the orthorhombic form are observable by Raman spectroscopy.

15.2.2 The Landau theory of continuous phase transitions

⁶ A phenomenological theory relates observations mathematically without tracing them back to a fundamental law.

In 1937, L. D. LANDAU introduced a phenomenological theory⁶ to treat continuous phase transitions in a uniform way. The theory has been considerably expanded by E. M. LIFSCHITZ and V. L. GINZBURG and later on by many more authors, and finally has been extended to cover discontinuous phase transitions as well [11, 264, 275–279].

First, we present the basic concepts of Landau theory for continuous phase transitions by means of the example of the tetragonal $\rightleftharpoons$ orthorhombic phase transition of calcium chloride. An order parameter is defined to trace the course of the phase transition. The *order parameter* is an appropriate, measurable quantity that is apt to account for the essential differences of the phases.

In the first instance, the order parameter is a quantity that can be measured macroscopically. Depending on the kind of phase transition, it can be, for example, the difference of the densities of phases 1 and 2 or the magnetization at a transition from a paramagnetic to a ferromagnetic phase. However, certain changing structural parameters can also be chosen as order parameters. The order parameter must fulfil certain conditions; these include:

In the lower-symmetry phase, the order parameter must change continuously with temperature (or pressure or some other variable of state) and finally it must vanish at the critical temperature T_c (or at the critical pressure p_c), i.e. it must become zero; in the higher-symmetry phase, after having surpassed T_c , it remains zero. The *critical temperature* corresponds to the point of transition (at a temperature-dependent transition). For transitions in the solid state, the order parameter has to fulfil certain symmetry properties that are explained on the following pages. As explained in Appendix B (Section B.2, page 271), the order parameter η can be used to expand a power series for the Gibbs free energy, $G = G_0 + \frac{1}{2}a_2\eta^2 + \frac{1}{4}a_4\eta^4 + \frac{1}{6}a_6\eta^6 + \dots$. This power series comprises only even powers of η , i.e. it is invariant to changes of the sign of η .

CaCl_2 is tetragonal in the rutile type above the transition temperature $T_c = 490 \text{ K}$ ($P4_2/mnm$); below it is orthorhombic in the CaCl_2 type (Figs. 1.2 and 1.3, page 6). As compared to the tetragonal form, the strands of edge-sharing coordination octahedra running parallel to **c** have been mutually rotated in the orthorhombic form. The angle of rotation becomes larger the further the temperature is below T_c . As with all (static) structural descriptions, the stated position of the octahedra refers to their mean positions, i.e. the equilibrium positions of the atomic vibrations.

Even with the slightest rotation of the octahedra the symmetry can no longer be tetragonal. The lattice parameters a and b become unequal and the mirror planes running diagonally through the unit cell of the rutile structure cannot be retained (in addition to some more symmetry elements). The symmetry of the CaCl_2 type has to be that orthorhombic subgroup of $P4_2/mnm$ in which these mirror planes have been omitted; that is the space group $P2_1/n2_1/n2/m$ ($Pnnm$). Universally it holds:

First criterion of Landau theory: At a continuous phase transition, the space group $\mathcal{H}$ of one phase needs to be a subgroup of the space group $\mathcal{G}$ of the other phase: $\mathcal{H} < \mathcal{G}$ ('the symmetry is broken'). It does not have to be a maximal subgroup.

Starting at a temperature $T > T_c = 490$ K, upon reduction of temperature, the frequency of the B_{1g} mode of the octahedra of tetragonal CaCl_2 shifts towards zero. This kind of behaviour is called a *soft mode*. After the temperature has fallen below the critical temperature $T_c = 490$ K, the equilibrium position of the octahedra has switched from that of the tetragonal to that of the orthorhombic form. The strands of octahedra now perform rotational vibration about the new equilibrium position, which now corresponds to the symmetry species A_g of space group $Pnnm$. The frequency shift is reversed, with rising frequencies the more the temperature is below T_c .

Since the motion of atoms during the phase transition is directly connected with the vibrational mode, one says, 'the phase transition is driven by a soft mode' or 'the mode condenses' or 'the mode freezes'. The symmetry species of the higher-symmetry phase is called the *active representation*.

Second criterion of Landau theory: The symmetry breach is due to only one irreducible representation (symmetry species), the active representation. In the higher-symmetry phase it may not be the identity representation (totally symmetrical vibration). Upon the phase transition it becomes the identity representation of the lower-symmetry phase.

In Fig. 15.3 the vector $\vec{\eta}$ has been drawn that marks the shift of the equilibrium position of a Cl atom as compared to the position in the high-temperature form. The vector shows how the coordination octahedra have been rotated. At a given temperature $T < T_c$ the octahedra have been rotated by a certain angle, as compared to the rutile type, and the vector $\vec{\eta}$ has some specific length η . Rotation of the octahedra in the opposite direction $-\vec{\eta}$ is completely equivalent energetically. In the tetragonal high-temperature modification the value is $\eta = 0$. The value of η of the shift can be used as an order parameter.

According to the theory, the order parameter must behave with respect to symmetry in the same way as the active representation (it must be 'transformed like the active representation'); for CaCl_2 this is the symmetry species B_{1g} of $P4_2/mnm$. As can be checked in corresponding tables (in the 'character table' of the point group $4/mmm$, see, e.g. [15, 16, 22]), a 'basis function' $x^2 - y^2$ belongs to B_{1g} , x and y being Cartesian coordinates. In our case, the coordinates x and y of the Cl atom can be taken. η being proportional to $x - y$, and with $x^2 - y^2 = (x - y)(x + y)$ and $x + y \approx \text{constant}$, η is proportional to $x^2 - y^2$ to a good approximation and is thus an appropriate order parameter. A couple of

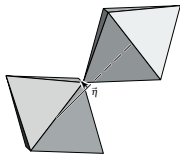


Fig. 15.3 Two connected octahedra in the low-temperature modification of CaCl_2 and shift vector $\vec{\eta}$ of the common Cl atom as compared to the position in the high-temperature modification.

further variables can be used as approximate order parameters for CaCl_2 ; due to $\sin\varphi \approx \tan\varphi \approx \varphi$ this is valid for the (small) angle of rotation φ of the octahedra and for the ratio $(b-a)/(b+a)$ of the lattice parameters (the so-called spontaneous deformation).

In the lower-symmetry phase ($T < T_c$) the order parameter follows a power law as mentioned in the margin. A is a constant and β is the *critical exponent*. As explained in Appendix B, $\beta \approx 0.5$ holds if there are long-range interactions between the particles; this is fulfilled for CaCl_2 , due to the interconnection of all octahedra. For short ranges of interaction, for example, for magnetic interactions, $\beta \approx 0.33$.

Landau theory agrees well with the experimental data shown in Fig. 15.4. There is no discontinuous change of volume at the point of transition. The order parameter $\eta' = (b-a)/(b+a)$ is consistent with the power law (15.2) over a large temperature range, a critical exponent of $\beta = 0.45$ matching the experimental dependence quite well. One lattice mode of symmetry species

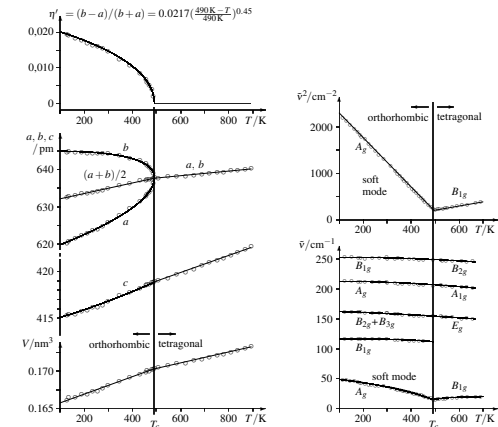


Fig. 15.4 Upper left: Plot of the temperature-dependent course of the order parameter η' of CaCl_2 . Lower left: Temperature dependence of the lattice parameters and the unit cell volume of CaCl_2 [38, 40]. Right: Temperature dependence of the frequencies of Raman-active vibrations of CaCl_2 [280]. The B_{1g} mode of the orthorhombic modification at 120 cm^{-1} changes to the symmetry species A_{2g} of the tetragonal modification and becomes Raman-inactive.

B_{1g} of $P4_2/mnm$ exhibits soft mode behaviour and switches to the symmetry species A_g of $Pnnm$, while the frequencies of all other modes remain nearly temperature independent, even across the point of transition. The minimum of the frequency of the soft mode is exactly at the transition temperature found by crystallography.

The same kind of rutile type $\rightleftharpoons$ CaCl_2 type phase transition has been observed with further compounds (e.g. CaBr_2 , MgF_2 , NiF_2 , ZnF_2 , stishovite- SiO_2 , SnO_2). If the phase transition is driven by pressure, the CaCl_2 type is the high-pressure modification.

The sketched model of a continuous phase transition, driven by a lattice mode at the Γ point of the Brillouin zone, matches the notion of a continuous change from one structure to the other with the synchronous participation of all atoms of the crystal (or of a large domain of the crystal). A change of space group takes place at the point of transition. In addition, certain physical properties change abruptly. For example, CaCl_2 becomes ferroelastic below the point of conversion ('pure and proper ferroelastic' [281]).

If a lattice mode at the Γ point is involved, like in the case of the rutile type $\rightleftharpoons$ CaCl_2 type transformation, the space groups of both phases are *translationengleiche* (their primitive unit cells have the same size). If the freezing mode is at a boundary point of the Brillouin zone, atoms in unit cells adjacent in the corresponding direction vibrate with exactly opposite displacements; after the mode has frozen, the volume of the unit cell has been doubled. This means that the symmetry breach must involve a step of symmetry reduction with a *klassengleiche* group-subgroup relation of index 2.

The symbols listed in Table C.1 in Appendix C are not applicable to lattice vibrations outside the Γ point, i.e. for modes that have atomic motions that differ from unit cell to unit cell. For this case, other symbols are used [282, 283].

If a non-totally symmetrical lattice mode freezes, the result is a distorted structure whose space group is a subgroup of the original space group. Such a subgroup is called an *isotropy subgroup*. Which isotropy subgroups occur depending on symmetry species has been listed for all symmetry species of all space groups [283]. They can also be determined with the computer program ISOTROPY [284]. For the theoretically important interplay between lattice modes and isotropy subgroups refer to, for example, [275, 285, 286].

15.3 Domains and twinned crystals

A phase transitions in the solid state often results in a *domain structure* of crystalline phases. The domain structure is the result of nucleation and growth processes. If the arrangement of the building blocks in both phases is rather different, there is no crystallographic relation between the orientations of the initial and the new phase. The nuclei form with orientations at random, depending on the energetic conditions at the defect sites in the crystal. This case is frequently encountered among molecular crystals.

However, if the lattices of the two crystalline phases match to some degree, for a nucleus of the new structure that develops at a defect site of the old

structure, it is more favourable to keep the orientation of the old phase. The orientational relations between the phases before and after the transformation, as a rule, are not the result of a homogeneous process involving a simultaneous motion of all atoms in a single crystal. The crystalline matrix of the substrate rather governs the preferred orientation adopted by the nuclei that are formed during the course of the nucleation process. The crystallites that result from the subsequent growth of the nuclei maintain their orientations.

Even at a continuous phase transition that is driven by a soft mode, as a rule, it is not the whole crystal that transforms at once. The ever existing faults in a crystal and the mosaic structure of real crystals impede lattice modes running uniformly through the whole crystal, and the gradients that inevitably emerge in a crystal during a temperature or pressure change cause different conditions in different regions of the crystal. In addition, fluctuations arise when the point of transition is approached; they differ statistically in different regions of the crystal (cf. Appendix B, Section B.3, for the fluctuations).

Therefore, the phase transition starts in different regions of the crystal simultaneously or at consecutive time intervals, such that domains result. The domains grow until they meet. If two matching domains meet, they can combine to form a larger domain; otherwise a domain boundary results between them. Once formed domain boundaries can advance through the crystal, one of the domains growing at the expense of the other one.

The resulting system of intergrown crystallites is called a *topotactic texture* after W. KLEBER [287].

Under these circumstances, aspect 3 of the symmetry principle, as stated in Section 1.1, is fully effective [12, 106, 289, 290]. A phase transition that is connected with a symmetry reduction will result in new phases that consist of

twin domains

if the formed phase belongs to a crystal class of reduced symmetry, or

antiphase domains,

if translational symmetry is lost.

Therefore, twinned crystals are formed if the new phase belongs to a *translationengleiche* subgroup, and antiphase domains if it is a *klassengleiche* subgroup. In the physical literature, phase transitions between *translationengleiche* space groups are called *ferroic transitions*, the lower-symmetry phase being the ferroic phase and the higher-symmetry phase the *para* phase. Non-ferroic transitions take place between *klassengleiche* space groups. For stricter definitions of these terms see [288].

Twin domains are dealt with in Section 15.5, antiphase domains in Section 15.6.

The total number of domains formed, of course, depends on the number of nucleation sites. The number of different *domain kinds*, however, is ruled by the index of the symmetry reduction. At a *translationengleiche* symmetry reduction of index 3 (t3 group-subgroup relation) we can expect twins with three kinds of domains, having three different orientations. An isomorphic subgroup of index 5 (i5 relation), since it is a *klassengleiche* symmetry reduction, will entail five kinds of antiphase domains. If the symmetry reduction includes several steps (in a chain of several maximal subgroups), the domain structure

will become more complicated. With two t_2 group-subgroup relations, we can expect twins of twins.

The actual number of observable domain kinds may be less than expected if a domain kind is not formed during nucleation. This can be controlled by the experimental conditions. For example, an external electric field can suppress the formation of more than one kind of differently oriented ferroelectric domain at a phase transition from a paraelectric to a ferroelectric modification. Among very small crystals (in the nanometre range), it can happen that only one nucleus is formed per crystal and that its growth is fast enough to fill the whole crystal before a second nucleus is formed, with the result of single-domain crystals [291].

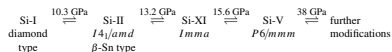
As a rule, among temperature-driven phase transitions, the high-temperature form has the higher symmetry. Among phase transitions induced by electric or magnetic fields, the lower-symmetry phase is the one persisting while the field is being applied. No corresponding rules can be stated for phase transitions induced by pressure; under pressure, the more dense phase is the more stable one.

The symmetry relations are also valid for *topotactic reactions*, when a chemical reaction takes place in the crystalline solid and the orientation of the domains of the product is determined by the original crystal (Section 16.1).

15.4 Can a reconstructive phase transition proceed via a common subgroup?

Some substances exhibit two subsequent *displacive* phase transitions (at two different temperatures or pressures) with small atomic displacements, such that the first transition involves a symmetry reduction, the second one a symmetry enhancement.

Silicon offers an example. When increasing pressures are exerted upon silicon, it experiences several phase transitions:



Silicon-V ($P6_3/m2/m2/m$) has a simple hexagonal structure [292]. There exists no group-subgroup relation between $I4_1/a2/m2/d$ (β -tin type) and $P6_3/m2/m2/m$ and they have no common supergroup. However, the structure of Si-XI is intimately related with both that of Si-II and Si-V, and the space group of Si-XI, $I2_1/m2_1/m2_1/a$, is a common subgroup of $I4_1/a2/m2/d$ and $P6_3/m2/m2/m$ (Fig. 15.5).

Assuming no atomic displacements, the calculated coordinates of a Si atom of Si-XI would be $0, \frac{1}{4}, -0.125$ when derived from Si-II, and $0, \frac{1}{4}, 0.0$ when derived from Si-V. The actual coordinates are halfway between. The metric deviations of the lattices are small. The lattice parameter c of the hexagonal structure is approximately half the value of a of tetragonal Si-II. Taking into account the basis transformations given in Fig. 15.5, the expected lattice para-

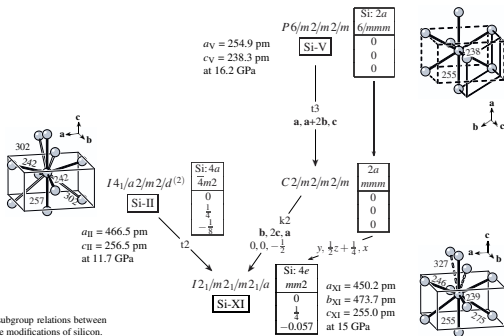
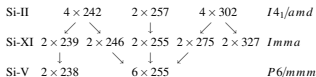


Fig. 15.5 Group-subgroup relations between three high-pressure modifications of silicon.

meters for Si-XI, calculated from those of Si-V, would be $a_{XI} = a_V \sqrt{3} = 441.5$ pm, $b_{XI} = 2c_V = 476.6$ pm, and $c_{XI} = a_V = 254.9$ pm. Compare this with the observed values (Fig. 15.5).

The coordination of a Si atom is (contact distances in pm up to 340 pm):



At the points of transition, volume discontinuities were observed experimentally, amounting to 0.2% for Si-II $\rightarrow$ Si-XI and to 0.5% for Si-XI $\rightarrow$ Si-V, and the atomic coordinates shift abruptly [292]. Therefore, the transitions are first-order transitions. However, the abrupt changes are small, and the model of two displacive phase transitions with small atom shifts seems plausible. Most notably, two separate phase transitions are actually observed. In a certain pressure range, the whole crystal actually consists of stable Si-XI. In this case, a group-theoretical relation between Si-II and Si-V via the common subgroup of Si-XI is justified.

A completely different situation is that of *reconstructive* phase transitions when there is no group-subgroup relation between the involved space groups. For such cases, the success of the theory of continuous phase transitions has misled researchers to conceive of mechanisms according to which two subsequent transformations take place via an intermediate phase, similar to the case

of the transformations of Si-II to Si-V via Si-XI. The hypothetical intermediate phase is assumed to have a space group that is a common subgroup of the initial and the final space groups.

Reconstructive phase transitions are always first-order transitions and show hysteresis. Therefore, a synchronous motion of atoms in the whole crystal is excluded. The transformation can proceed only by nucleation and growth, with coexistence of both phases during the transformation. Between the growing new and the receding old phase there are phase boundaries. The reconstruction of the structure occurs at and only at these boundaries. Any intermediate state is restricted to this interface between the two phases. The situation is fundamentally different from that of silicon where the existence of Si-XI is not restricted to an interface.

The space groups on either side of an interface are different. They cannot be mapped one onto the other by a symmetry operation; therefore, no symmetry element can exist at the interface. In particular, an intermediate phase whose existence would be restricted to the region of the interface cannot have any space group whatsoever. There is an additional aspect: Space groups describe some static state. During a snapshot of a few femtoseconds duration, no crystal fulfils a space group because the atoms are vibrating and nearly all of them are displaced from their equilibrium positions. Mean atomic positions that fulfil a space group can only be discerned after a somewhat longer time interval has elapsed. During a phase transition, the interface advances through the crystal and the atoms are in motion; there exist no mean atomic positions.

Devised reaction mechanisms for reconstructive phase transitions via hypothetical common subgroups are disconnected from reality. Approximately 10 different mechanisms were published merely for the transformation of the NaCl type to the CsCl type, with different common subgroups (see, e.g. [293–295] and references therein). The very number is suspect. Formally, arbitrarily many such mechanisms can be devised, as there are always an infinite number of common subgroups.

In such papers, detailed paths of motion of the atoms have been described, depicted in one unit cell. Tacitly, a synchronous motion of the atoms in all unit cells of the crystal is thus suggested, proceeding until they snap into place in the assumed intermediate space group, followed by a synchronous motion from the intermediate phase to the final phase. Synchronous motion, however, is excluded for a first-order transition. But perhaps it has not been meant to be this way, but tacitly only a shift of atoms in one cell has been assumed, followed by further cells, like in a row of falling dominoes. This would be a realistic model with nucleation and subsequent growth. However: this does not require and is not in accordance with the occurrence of a hypothetical intermediate phase. It may well be that, for a few femtoseconds, the atoms are arranged in a couple of cells as in one cell of the assumed intermediate phase. However, it is impossible to assign a space group to this state.

15.5 Growth and transformation twins

Definition 15.4 An intergrowth of two or more macroscopic, congruent, or enantiomorphic individuals of the same crystal species is a twin, if there is a crystal-symmetric relative orientation between the individuals [297].

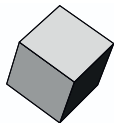
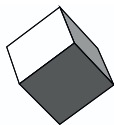


Fig. 15.6 Centre: View of a twin of the mineral fluorite having the twin axis $[111]$. The two cubes are mutually rotated about the space diagonal $[111]$ by exactly 180° .

The individuals are the twin components or twin domains. The *twin law* specifies their mutual orientation. The twin law is a symmetry operation that maps one domain onto the other one. The twin element is the corresponding symmetry element. This symmetry operation does not belong to the point group of the crystal.

The domain boundary or twin interface is a planar or irregular surface at which the twin components are in contact. If the structures of the adjacent domains have to be harmonized by certain distortions on either side of the boundary, the domain boundary is in the midst of a domain wall that can have a thickness of up to a few unit cells. If the domain boundary is planar, on the atomic scale, its symmetry is a layer group.

The occurrence of twins is a widespread phenomenon. One has to distinguish whether the twins have developed during the growth of the crystals or by a phase transition in the solid state.

Among **growth twins**, the formation of nuclei of crystallization decides how the individuals are intergrown. In this case, group-subgroup relations between space groups are irrelevant.

The shape of a kind of the frequently occurring twins of the cubic mineral fluorite (CaF_2) is shown in Fig. 15.6. This is an example of a *penetration twin*; these occur only among growth twins. Starting from a nucleus, the crystal grows with two different orientations that are mutually rotated by 180° about $[111]$. Actually, the twin consists of twelve pyramid-shaped domains, six of each orientation, originating from a common point in the centre of the crystal. Contact twins are simpler; the twin components have a common a plane. A polysynthetic (or lamellar) twin is formed by repetition of contact twins and consists of a sequence of domains with two alternating orientation states.

Transformation twins are formed during a phase transition in the solid state. Frequently, there exists a group-subgroup relation between the involved space groups and the kind of twinning depends on this relation. The group-subgroup relation between the two modifications of calcium chloride is shown in Fig. 1.2 (page 6). $Pnnm$ is a *translationengleiche* subgroup of index 2 of $P4_2/mnm$, so we can expect the formation of twins with two orientational states when CaCl_2 is transformed from the tetragonal rutile type to the orthorhombic CaCl_2 type. The lattice parameters a and b , being equal in the tetragonal structure, become unequal during the phase transition, either $a < b$ or $a > b$. If we retain the directions of the basis vectors $\mathbf{a}$ and $\mathbf{b}$, then one of the orientational states of the twin components is that with $a < b$, the other one that with $a > b$ (Fig. 1.3, page 6). The two components can be mapped one onto the other by a symmetry operation that has disappeared at the symmetry reduction, for example by reflection through a plane that runs diagonally through the unit cell of the rutile type. This (former) symmetry element becomes the twin element.

During the phase transition, the distinction of the lattice parameters taking place within the matrix of the tetragonal crystal causes mechanical stress. The occurrence of differently oriented twin domains partly alleviates the stress. Therefore, a crystal consisting of many twin domains will be energetically favoured as compared to a single-domain crystal (Fig. 15.7). A uniformly synchronous motion of all atoms in the whole crystal resulting in a single-domain crystal is less probable.

Potassium sulfate crystallizes from aqueous solution in an orthorhombic modification with a pseudo-hexagonal structure [298, 299]:

$$Pmcn, a = 576.3 \text{ pm}, b = 1007.1 \text{ pm} (\approx a\sqrt{3} = 998.2 \text{ pm}), c = 747.6 \text{ pm}$$

Growth twins are formed (depending on crystallization conditions) adopting the shape of pseudo-hexagonal plates that are subdivided in three pairs of sector domains with 60° angles (Fig. 15.7). Upon heating, a first-order phase transformation takes place at 587°C , at which the domains coalesce to one domain, the crystal becoming hexagonal ($P6_3/mmc$). In the hexagonal modification the SO_4^{2-} ions are misordered in at least two orientations (Fig. 15.8).

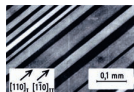
When allowed to cool again, the SO_4^{2-} ions regain their order, and again orthorhombic twins with three kinds of domains are formed, but this time as transformation twins with lamellar domains.

Isotopic ammonium sulfate also forms the same kind of growth twins. By slight compression, the domain boundaries can be caused to advance through the crystal until a single-domain single crystal is obtained [303]. The expected phase transition to a hexagonal high-temperature form does not occur because ammonium sulfate melts 'prematurely'. Therefore, in this case, the hexagonal aristotype is a hypothetical modification ('paraphase', 'prototype phase') [304]. In contrast to potassium sulfate, ammonium sulfate exhibits a phase transition to a ferroelectric modification below -50°C in which the SO_4^{2-} tetrahedrons are tilted. The corresponding group-subgroup relations are shown in Fig. 15.8.

15.6 Antiphase domains

The phase transition of Cu_3Au from its misordered high-temperature phase in the space group $Fm\bar{3}m$ to the ordered phase involves a *klassengleiche* symmetry reduction of index 4 to the space group $Pm\bar{3}m$ (Fig. 1.4, page 7). Therefore, the occurrence of antiphase domains can be expected. The crystal class is retained, and the size of the primitive unit cell is enlarged by a factor of 4 (because the face-centring disappears).

Consider a nucleus of crystallization of the ordered phase and its growth, and let the gold atoms occupy the vertices of the initial unit cell. Let another nucleus be formed at some other site in the crystal, but with a shifted origin, i.e. with a gold atom at one of the face centres of the initial unit cell. Somewhere the growing domains will meet. Even though the unit cells have the same size and the same orientation, they do not match with one another because their unit cells are displaced by half of a face diagonal. A domain boundary emerges at which the 'wrong' atoms come to be next to each other. This is an antiphase



CaBr₂



K₂SO₄

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Fig. 15.7 Top: Transformation twin with domains I and II of a crystal of CaBr_2 (CaCl_2 type); direction of view along c , photograph taken with crossed polarization filters [296]. Bottom: Growth twin of K_2SO_4 having six domains, precipitated from aqueous solution. View onto (001) of a plate, about 1 mm thick and 5 mm in diameter (taken between polarization filters to enhance the contrast [297]; M. Moret, Milano).

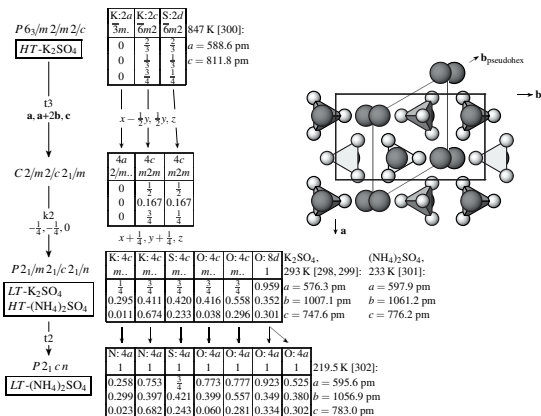


Fig. 15.8 Crystal structure of the low-temperature form of potassium sulfate and group-subgroup relations from the hexagonal high-temperature form, in which the sulfate ions are misordered (for this reason no O atom coordinates are given). Ammonium sulfate forms a non-centrosymmetric, ferroelectric low-temperature form.

boundary between antiphase domains (they have shifted phases). Since the gold atoms have the choice to occupy one out of four positions $(0, 0, 0, \frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, 0, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2})$ of the initial cell, four kinds of domains result. Antiphase domains are also called translation twins or translation domains.

Antiphase domains do not show up in X-ray diffraction; the structure determination is not hampered (except when the domains are very small and part of the Bragg reflections spread to diffuse scattering). Antiphase boundaries can be made visible by electron microscopy.

Consider the example of the structures of the high- and low-temperature modifications of $BaAl_2O_4$ shown in Fig. 15.9. These are stuffed tridymite structures, with a framework of vertex-sharing AlO_4 tetrahedra and Ba^{2+} ions in hexagonal channels. At high temperatures $BaAl_2O_4$ is paraelectric in the space group $P6_322$. Upon cooling, a phase transition takes place between 670 and 400 K, resulting in a ferroelectric modification in the space group $P6_3$, with the AlO_4 tetrahedra being mutually tilted and with doubled lattice parameters a and b . This requires the two steps of symmetry reduction shown in the margin.

The *translationengleiche* step causes the occurrence of twin domains that are distinct by opposite orientations (opposite c directions) and that alternate

$P6_322$
 $\downarrow t_2$
 $P6_3$
 $\downarrow i_4$
 $2a, 2b, c$
 $\downarrow$
 $P6_3$

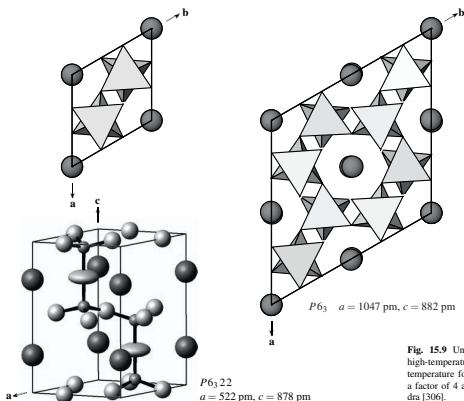


Fig. 15.9 Unit cells of BaAl_2O_4 . Left: high-temperature form [305]. Right: low-temperature form with unit cell enlarged by a factor of 4 and tilted coordination tetrahedra [306].

in lamellae along c (Fig. 15.10 left). In addition, due to the isomorphic subgroup of index 4, four kinds of antiphase domains are observed. Figure 15.11 shows schematically how the unit cells of the four kinds of domains are mutually displaced. The domain boundaries can be discerned with an electron microscope (Fig. 15.10 centre and right).

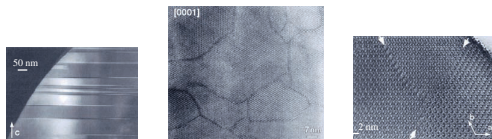


Fig. 15.10 Transmission electron-microscopic frames of BaAl_2O_4 [307]. Left: View along $[100]$ showing twin domains. Centre and right: View along $[001]$; arrows mark antiphase boundaries.

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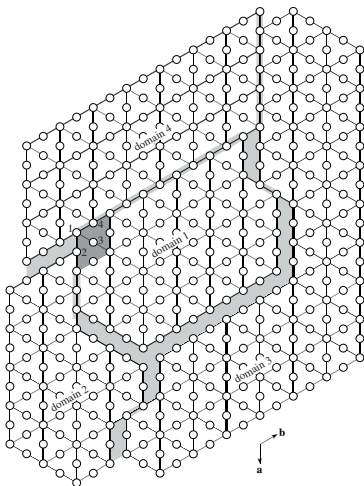


Fig. 15.11 Schematic drawing with the four kinds of antiphase domains of BaAl_2O_4 [307]. The circles represent the Ba atoms. Domain boundaries are enhanced in grey. One unit cell is marked in domain 1. The numbers at the Ba atoms designate the four different origin positions of the four domains.

Exercises

Solutions in Appendix D (page 296)

- (15.1) What are experimentally observable properties that can be used to decide whether a phase transition is of second order (after EHRENFEST) or if it is continuous?
- (15.2) May or must an 'isosymmetric' phase transition be a continuous transition?
- (15.3) In Fig. 15.4 (left) the curve for the cell volume V has a kink at T_c . May this be the case for a second-order phase transition?
- (15.4) The packing of several modifications of P_4S_3 and similar cage-like molecules is shown in Fig. 14.4 (page 190), and the corresponding symmetry relations are given in Fig. 14.3. All modifications have the same packing pattern, but with differently rotated molecules. Could these modifications be transformed one to the other by continuous phase transitions?
- (15.5) At high temperatures, BaTiO_3 has the cubic perovskite structure, space group $Pm\bar{3}m$. Upon cooling it is distorted, adopting the space group $P4mm$. Will the crys-

tals of the low-symmetry structure be twinned? If so, with how many kinds of domains?

- (15.6) SrTiO_3 has the cubic perovskite structure ($Pm\bar{3}m$). Upon cooling below 105 K, the coordination octahedra are mutually rotated and the symmetry is reduced to $I4/mcm$, including an enlargement of the unit cell by a factor of 4 ($a' = b' = a\sqrt{2}$, $c' = 2c$). Will the crystals of the low-symmetry structure be twinned? If so, with how many kinds of domains?
- (15.7) $\text{SrCu}_2(\text{BO}_3)_2$ exhibits a second-order phase transition at 395 K. Above 395 K its space group is $I4/mcm$, below it is $I\bar{4}2m$ with the same lattice parameters [308]. Upon cooling beyond the point of transition, can one expect the formation of twins? If so, will these cause problems for X-ray crystal structure determination?
- (15.8) The structure of ErCo_2 experiences a distortion during the phase transition from the paramagnetic to the ferrimagnetic modification at 32 K. At $T > 32$ K it is cubic, space group $Fd\bar{3}m$, $a = 713.3$ pm; below 32 K it is rhombohedral, space group $F\bar{3}m$, $a = 714.5$ pm, $\alpha = 89.91^\circ$ (non-conventional face-centred setting for $R\bar{3}m$ with rhombohedral axes) [309]. Why do the X-ray reflections of the ferrimagnetic phase show a broadening or splitting?
- (15.9) According to quantum-mechanical calculations, sillimanite, Al_2SiO_5 , should exhibit a phase transition if

pressure is exerted to more than 30 GPa. The atoms are expected to move together, with an increase of the coordination number of the Si atoms from 4 to 5. The calculated change of the corresponding Si–O distances and of the lattice parameter a as a function of pressure is shown below (arrows mark the directions of pressure increase and decrease) [310]. The remaining lattice parameters and the structure in general are changed only marginally. The space group is $Pnma$ before and after the transition. Is this a continuous phase transition? Is the transformation reconstructive or displacive?

